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Prepared by

1.	LEE YIN SHEN	(A23CS0236)
:	2. ELIJAH SHE YU SHENG	(A23CS0073)
	3. NEO LI XIN	(A23CS0253)

Section : 02

Lecturer : Dr Rozilawati binti Dollah @ Md Zain

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1.0 Introduction

For this assignment, we are tasked to collect and analyse the dataset chose to identify the potential business insights. We collected a dataset that represents the customer's shopping behaviour in the fashion industry. It provides comprehensive information about customer's product preferences and spending habits during their shopping experience. This dataset encompasses a range of variables, from customers profiling and clothing categories to payment details. We hope that through this dataset, we can clearly showcase the customers' shopping preferences with data visualization, thus aiding the company in identifying the market trends to increase the business sales and profits.

1.1 Overview of the dataset

The dataset mainly focuses on the customers' spending behaviour in the fashion industry. The dataset is sourced from Kaggle, a data science platform which contains various types of datasets. It was first published by Shahzad Aslam and the dataset is scheduled to be updated annually. This dataset contains 3900 rows and 21 columns. The dataset is fully completed, and no null or duplicate data are detected. Below is the list of data that are included in this dataset:

No.	Feature Name	No.	Feature Name
1	Customer ID	12	Subscription Status
2	Age	13	Shipping Type
3	Gender	14	Discount Applied
4	Item Purchased	15	Promo Code Used
5	Category	16	Previous Purchases
6	Purchase Amount (USD)	17	Payment Method
7	Location	18	Frequency of Purchases
8	Size	19	Discount Percentage
9	Colour	20	Customer Income Level
10	Season	21	Item Price
11	Review Rating		

Table 1.1.1: Data Set List

1.2 Objective of the analysis

The primary goal of this analysis is to employ clustering models to understand and predict customer behaviour patterns. Specifically, the analysis aims to:

1. Classify customers based on age group dominance
Identifying which age groups are more prevalent in the customer base and understanding their characteristics.
2. Determine payment method preferences
Classifying customers based on their chosen method of payment and identifying demographic trends.
3. Assess the influence of income levels on purchasing behaviour
Investigating whether income brackets significantly influence how often or how much customers spend.

By building and evaluating multiple clustering models, the analysis will reveal underlying trends and relationships that can inform marketing strategies, product placement, and customer experience optimization.

1.3 Business insights to be explored

Understanding customer behaviour is critical to developing data-driven business strategies. The analysis will aim to extract the following actionable insights:

1. Dominant Customer Demographics
Identifying which age groups form the core of the customer base can help tailor products and marketing strategies more effectively.
2. Preferred Payment Channels
Understanding payment preferences can guide financial partnerships (e.g., with digital wallet providers), streamline checkout processes, and increase customer satisfaction.
3. Income vs. Spending Behaviour
Analysing the relationship between income and purchasing frequency or volume will help in segmenting customers by value and targeting high-value segments with loyalty programs or premium offerings.

These insights are crucial for businesses seeking to improve customer acquisition, retention, and overall satisfaction by aligning operational strategies with customer preferences and behaviours.

1.4 Clustering methods selected for the analysis

To carry out the clustering analysis, we selected two machine learning models: K-Means and DBSCAN. These algorithms were chosen for their ability to uncover patterns in the data and provide meaningful segmentation based on both numerical and categorical attributes.

1.4.1 K-Means Clustering

K-Means clustering is a widely-used partitioning method that divides data into a predefined number of clusters based on their proximity to the cluster centroids. It is known for its simplicity, speed, and effectiveness in handling large datasets. One of the main advantages of K-Means is its efficiency in grouping similar data points when the clusters are spherical and roughly equal in size. However, the limitation of K-Means is that it requires users to specify the number of clusters beforehand, which may not always be intuitive. Furthermore, it is sensitive to outliers and may not perform well when the clusters have irregular shapes or vary in density.

1.4.2 DBSCAN

DBSCAN, or Density-Based Spatial Clustering of Applications with Noise, is a density-based clustering algorithm that groups data points based on the density of their neighbourhoods. It does not require the number of clusters to be specified in advance and is capable of detecting clusters of arbitrary shape. This makes DBSCAN especially useful in identifying outliers and noisy data. Among its advantages are its flexibility and its effectiveness in dealing with datasets that contain noise or clusters of varying shapes. However, DBSCAN's performance relies heavily on the correct selection of parameters such as the neighbourhood's radius (Eps) and the minimum number of points (MinPts), and it may struggle with datasets containing clusters of varying densities.

2.0 Data Preparation

Before moving to the machine learning analysis, we do the data preprocessing to ensure accurate results. The data preparation was done using Python in Google Colab, where we applied techniques like normalization, encoding, and selecting important features. These steps are to prepare a dataset that is ready for analysis.

2.1 Data Normalization

```
scaler = MinMaxScaler()  
dataset[['Purchase Amount (USD)', 'Item Price']] = scaler.fit_transform(dataset[['Purchase Amount (USD)', 'Item Price']])
```

We normalize the Purchase Amount (USD) and Item Price columns using Min-Max scaling to bring them within the same range of [0, 1]. This ensures that no feature dominates the clustering process due to its scale, allowing the classification algorithms to treat all features equally and improve the accuracy of clustering.

2.2 Data Encoding

```
# Use LabelEncoder to convert categorical columns into numeric format  
label_encoder = LabelEncoder()  
  
# Encoding Gender, Category, Subscription Status, Location, Payment Method, etc.  
dataset['Gender'] = label_encoder.fit_transform(dataset['Gender'])  
dataset['Category'] = label_encoder.fit_transform(dataset['Category'])  
dataset['Subscription Status'] = label_encoder.fit_transform(dataset['Subscription Status'])  
dataset['Location'] = label_encoder.fit_transform(dataset['Location'])  
dataset['Payment Method'] = label_encoder.fit_transform(dataset['Payment Method'])  
dataset['Frequency of Purchases'] = label_encoder.fit_transform(dataset['Frequency of Purchases'])  
dataset['Customer Income Level'] = label_encoder.fit_transform(dataset['Customer Income Level'])
```

We encode the categorical columns like Gender, Category, Subscription Status, Location, Payment Method, Frequency of Purchases and Customer Income Level using Label Encoder. This converts each unique category into a numerical label, allowing these features to be used in the clustering model.

2.3 Data Export

```
output_file = 'preprocessed_data_full.csv'  
dataset.to_csv(output_file, index=False)
```

We export the finalized dataset into a new CSV file. This output file contains all selected and transformed attributes in a structured format, including headers and comma-separated values, making it suitable for subsequent model training and evaluation.

3.0 Clustering Models

This section assesses two clustering algorithms, K-Means and DBSCAN, to derive actionable insights from consumer data in the fashion industry. The performance of these clustering models is evaluated by analysing intra-cluster cohesion, inter-cluster separation, and cluster quality.

3.1 Age Group Dominance

Age is a key demographic variable that significantly influences shopping behaviour, product preferences, and spending patterns. The "Age" column in the dataset provides valuable insights into customer segmentation based on generational tendencies. By applying clustering models to this feature, we can uncover which age groups are more actively engaged in online fashion retail, identify shared traits among them, and support data-driven decisions for age-specific marketing and product recommendations.

3.1.1 K-Means Clustering

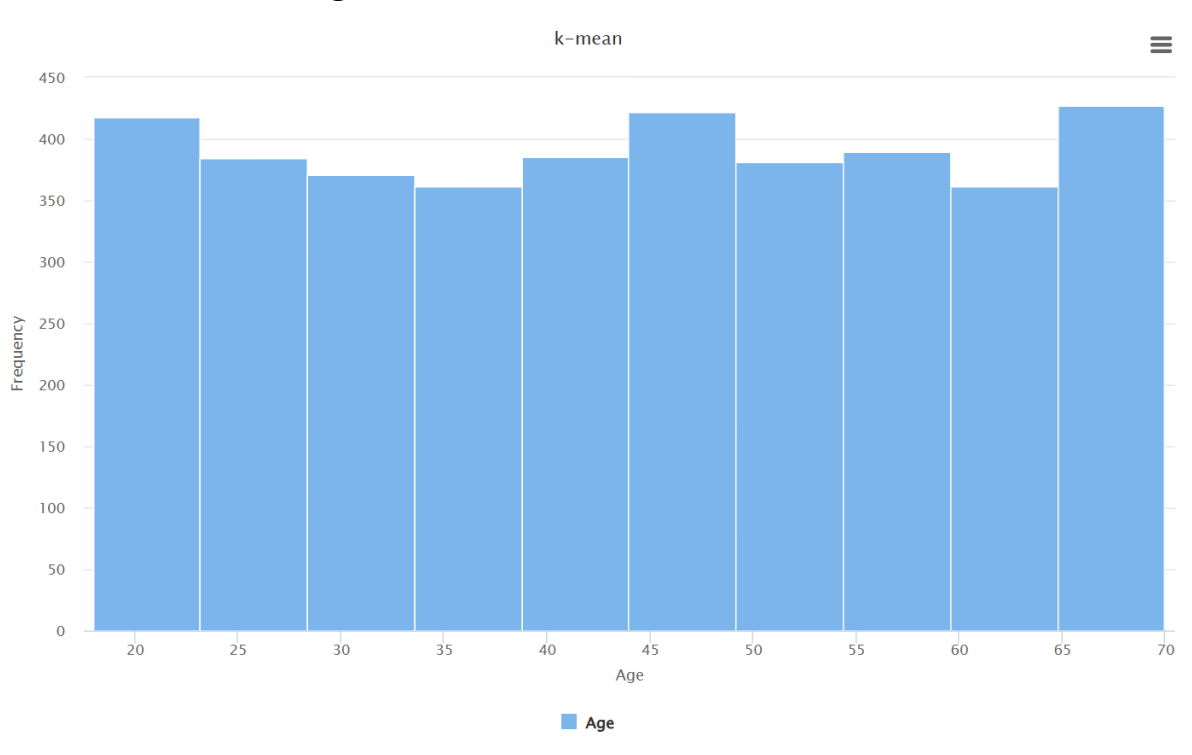


Figure 3.1.1.1 Histogram trained using K-Means Clustering

PerformanceVector

```
PerformanceVector:  
Avg. within centroid distance: 25.882  
Avg. within centroid distance_cluster_0: 27.107  
Avg. within centroid distance_cluster_1: 27.026  
Avg. within centroid distance_cluster_2: 23.346  
Davies Bouldin: 0.504
```

Using K-Means clustering, the dataset's "Age" column was segmented into three distinct clusters, revealing generational groupings such as younger adults which is Cluster 0, middle-aged consumers which is Cluster 1, and older adults which is Cluster 2. The clustering results show an average within-centroid distance of 25.882, with Cluster 2 being the most compact which is 23.346, and a low Davies-Bouldin index of 0.504, indicating well-separated and cohesive clusters. The histogram further supports a fairly uniform age distribution across the 18 to 70 range, with slight peaks at both ends. These insights support data-driven marketing strategies, enabling businesses to personalize campaigns, product recommendations, and engagement tactics for each age-based segment effectively.

3.1.2 DBSCAN

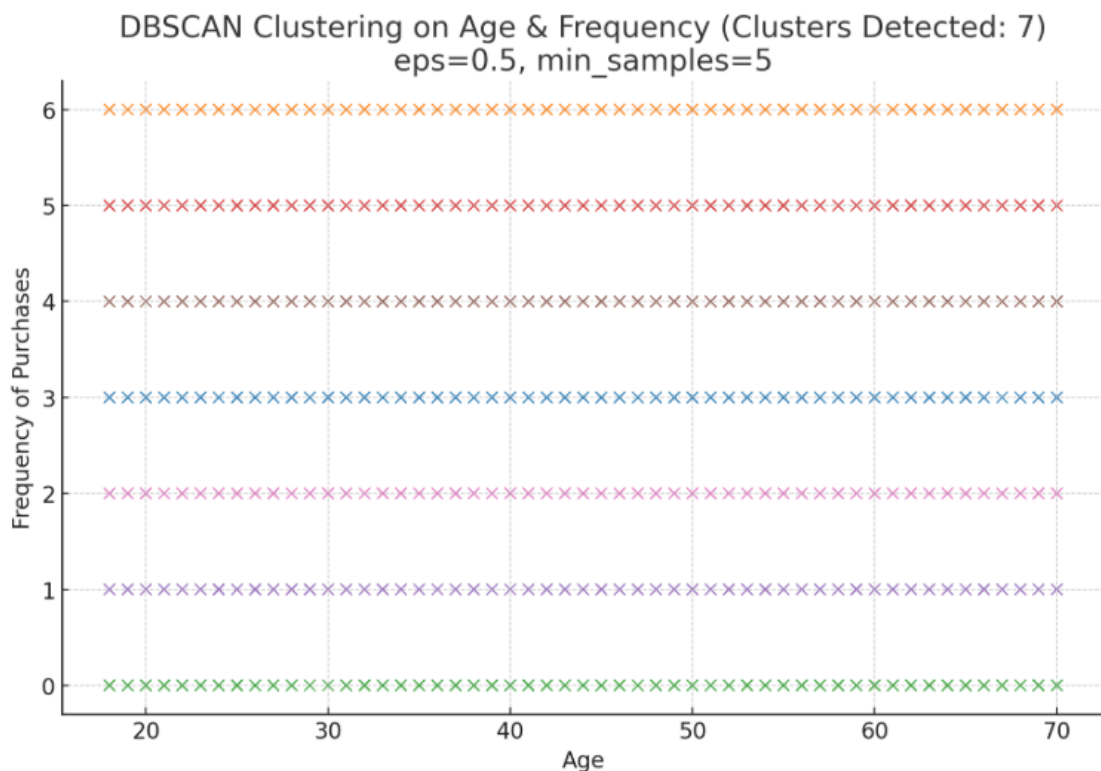


Figure 3.1.1.1 Scatterplot trained using DBSCAN Clustering

The scatter plot visualizes the results of DBSCAN clustering using two features: Age (x-axis) and Frequency of Purchases (y-axis). Each dot represents a customer, and colours indicate their assigned cluster. Notably, the points are aligned in horizontal rows, meaning customers with the same purchase frequency form distinct bands. The clustering results in 7 coloured bands, each corresponding to one of the 7 unique frequency values (0 through 6). The vertical spread (Age) within each band shows no clear separation by age, implying that DBSCAN formed clusters almost entirely based on frequency alone, not a combination of age and frequency.

Number of customers in each cluster (including noise):		
Cluster		
0	542	
1	539	
2	572	
3	563	
4	547	
5	553	
6	584	
Name: count, dtype: int64		
Average Age and Frequency per cluster (excluding noise):		
	Age	Frequency of Purchases
Cluster		
0	43.59	3.0
1	44.65	6.0
2	44.67	0.0
3	44.69	5.0
4	43.20	1.0
5	44.28	4.0
6	43.41	2.0
Number of noise points (label = -1):		
0		

Figure 3.1.1.1 RESULT trained using DBSCAN Clustering

The clustering output identified 7 distinct clusters, each corresponding to a unique purchase frequency level, with average ages ranging narrowly between 43 and 45 years. This indicates that while frequency of purchases varied across customers, age did not significantly influence clustering. Each cluster contains a balanced number of customers (around 540 and 580), and there were no noise points (outliers). Although the clustering is technically correct, it reveals that DBSCAN did not find complex patterns involving both age and frequency rather, it simply grouped users with identical frequency values together, which limits its usefulness for deeper customer segmentation.

3.2 Payment Method Preferences

Payment method is important transactional component that affects how people buy. The dataset's "Payment Method" column provides a useful perspective on how various payment methods impact consumer buying habits. We can see clear patterns in the relationship between chosen payment methods and item selection, order value, and purchase frequency by studying data.

3.2.1 K-Means Clustering

Cluster distribution by Payment Method:

Cluster	0	1	2
Payment Method			
Bank Transfer	196	218	198
Cash	211	227	232
Credit Card	218	234	219
Debit Card	216	219	201
PayPal	223	231	223
Venmo	219	220	195

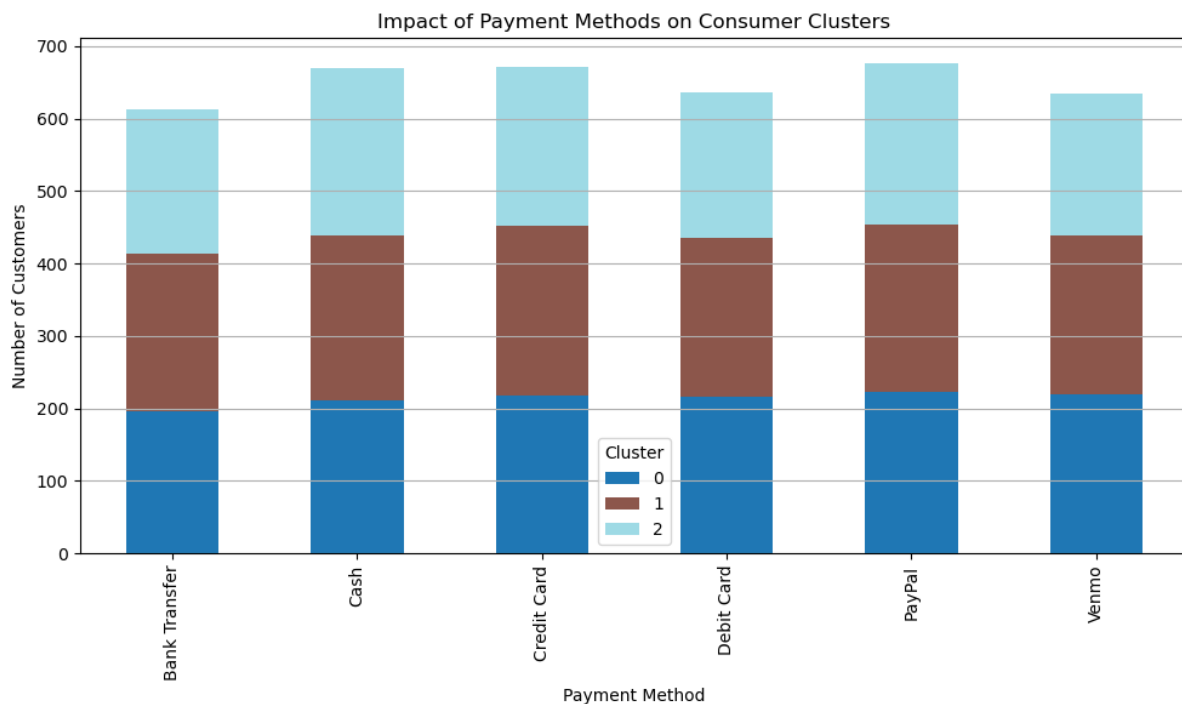


Figure 3.2.1.1 Bar Chart of K-Means clustering

The stacked bar chart, which is divided into three clusters, shows the relationship between various payment methods and consumer purchasing behaviour. Low-frequency, low-income customers are represented by Cluster 0, mid-frequency, higher-income buyers by Cluster 1, and high-frequency but lower-income buyers by Cluster 2. Although there are minor variations, the distribution is generally balanced across all payment types. Payment Method 4, for instance, has a little greater concentration of Cluster 1 clients, indicating that it is more well-liked by moderately wealthy consumers. In contrast, Payment Method 0 seems to draw a little more Cluster 0 users, suggesting that it would be in line with infrequent, frugal shoppers. Businesses might use this information to customise promotions or preferred payment methods according to the type of customer.

Cluster Interpretation Table:			
Cluster	Purchase Amount (USD)	Frequency of Purchases	Previous Purchases \
0	0.30	1.26	23.26
1	0.30	3.00	24.89
2	0.31	4.68	27.96

Customer Income Level	
Cluster	
0	0.48
1	2.00
2	0.51

Figure 3.2.1.1 Result trained using K-Means clustering

The Cluster Interpretation Table provides information into three different behaviour segments. With a low-income level of 0.48 and a low purchase frequency of just 1.26, Cluster 0 is made up of low-frequency, low-income buyers who may be more price-sensitive. With the greatest income level (2.00), a moderate frequency of purchases (3.00), and somewhat greater prior purchases, Cluster 1 is clearly the higher-income group. This probably indicates more stable, financially capable consumers who make regular but restrained purchases. With the highest frequency of purchases (4.68) and the greatest number of prior purchases (27.96), Cluster 2 exhibits the most active shopping behaviour; yet, their income level is comparatively low (0.51). This suggests a group of regular, devoted customers who may be frugal but always involved. Based on their behaviour and financial situation, these results assist in segmenting customers for more focused promotions.

3.2.2 DBSCAN

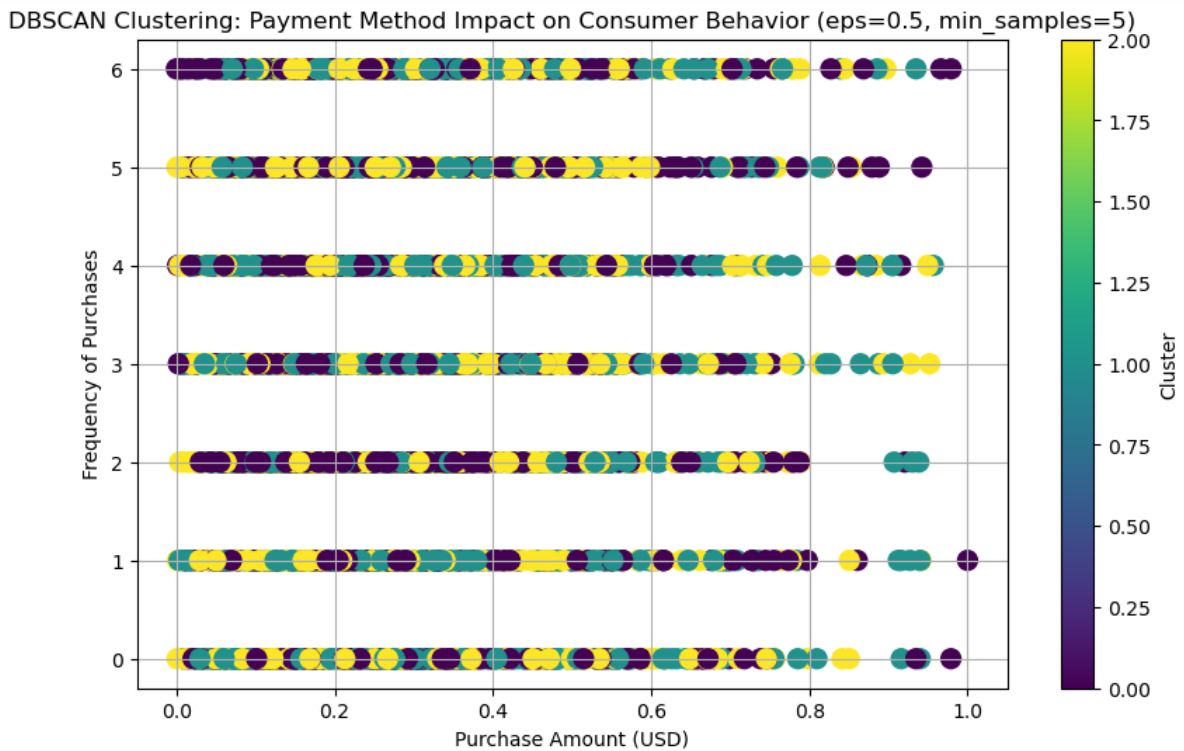


Figure 3.2.2.1 Scatter plot of DBSCAN clustering

The DBSCAN clustering visualization highlights how consumer behaviour, based on purchase amount and frequency, forms distinct groupings when analysed with parameters $\text{eps}=0.5$ and $\text{min_samples}=5$. Three major clusters (designated 0, 1, and 2) are visible in the plot, suggesting that customers frequently follow particular buying habits. For example, whilst some clusters comprise more diverse or sparse behaviours, others indicate high-frequency, low-amount purchases. The existence of several dense groupings raises the possibility that payment methods may have an impact on particular behavioural segments, maybe revealing consumer preferences or limitations. The dataset forms well-defined groupings under the current DBSCAN parameters, as evidenced by the successful cluster assignment of all data points and the absence of noise.

```

Number of points in each DBSCAN cluster (including noise):
DBSCAN Cluster
0    1349
2    1295
1    1256
Name: count, dtype: int64

Number of noise points: 0

Average values per cluster:
          Purchase Amount (USD)  Frequency of Purchases \
DBSCAN Cluster
0                0.30                3.00
1                0.31                2.98
2                0.30                2.94

          Previous Purchases  Customer Income Level
DBSCAN Cluster
0                24.89                2.0
1                25.66                1.0
2                25.53                0.0

DBSCAN Cluster distribution by Payment Method:
DBSCAN Cluster      0      1      2
Payment Method Name
Bank Transfer      218    195    199
Cash               227    214    229
Credit Card       234    211    226
Debit Card         219    210    207
PayPal             231    223    223
Venmo              220    203    211

```

Figure 3.3.2.2 Result trained using DBSCAN clustering

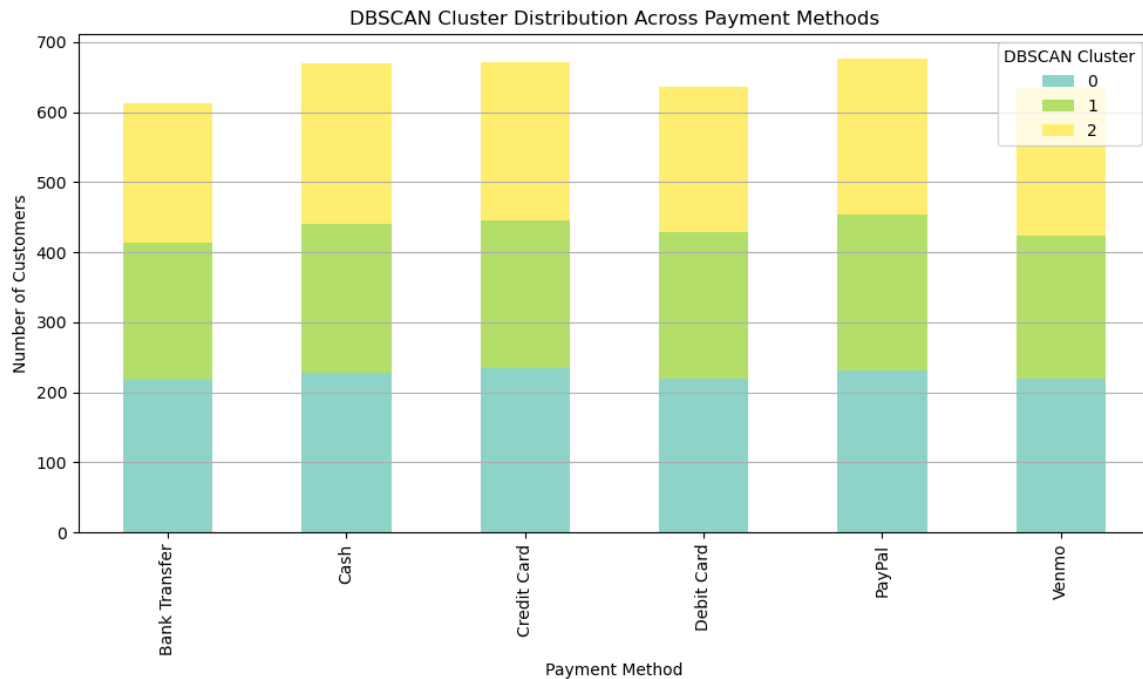


Figure 3.3.2.3 Bar Chart of DBSCAN clustering

With no noise points found, the DBSCAN clustering analysis separated the consumer data into three separate clusters (0, 1, and 2). This suggests that the data is well-structured for clustering. With 1,349 consumers in Cluster 0, 1,295 in Cluster 2, and 1,256 in Cluster 1, each cluster has a nicely balanced number of points.

Purchase behaviour appears to be consistent among clusters, as seen by the fact that, on average, all clusters exhibit extremely comparable purchasing behaviour in terms of purchase frequency (about three times) and purchase value (around \$0.30 to \$0.31). They vary considerably, though, in terms of prior purchases and client income levels:

- Cluster 0 has a middle-income level (2) and a moderate history of prior purchases (24.89).
- Cluster 1, which is associated with income level 1, had the most prior purchases (25.66).
- Cluster 2 is associated with the lowest income level (0) and has a significantly lower average of prior purchases (25.53).

The use of various payment methods is very evenly spread across all clusters when split down by cluster; no technique exhibits a clear preference for one over another. This suggests that other behavioural or demographic characteristics, such as income level and previous

shopping patterns, have a stronger influence on the clustering than payment method. All things considered, DBSCAN shows modest but significant segmentation in patterns of customer behaviour.

3.3 Impact of Customer Income Level on Purchasing Behaviour

Customer income level is crucial in determining purchasing behaviour. By analysing how income level relates to purchasing frequency, along with other factors like purchase amount, age, and previous purchases, we can gain a deeper understanding of consumer decision-making. This clustering analysis focuses on the relationship between income and purchasing frequency and some additional features to better understand customer purchasing patterns.

3.3.1 K-Means Clustering



Figure 3.3.1.1 Scatter plot of K-means clustering

From the scatter plot, we can observe the distribution of customer clusters based on Purchase Amount (USD) and Frequency of Purchases. The plot clearly illustrates how the clusters differ in terms of spending and purchase frequency. Cluster 0, represented by the purple colour, contains customers with low spending and infrequent purchases. Cluster 1, marked by green, groups customers with moderate spending and a moderate frequency of

purchases. Finally, Cluster 2, in yellow, represents customers with relatively higher spending and frequent purchases. The plot visually captures the distinct purchasing behaviours of each cluster, highlighting how customers' income levels influence their spending patterns.

Cluster Centers (Scaled Values):				
[[0.24007782	0.30445246	0.21102464	0.47604011	0.45665052]
[1.	0.29891551	0.50049419	0.50466157	0.48760231]
[0.25236967	0.30417337	0.77988415	0.52340807	0.54787697]]
Dataset with Cluster Labels:				
	Customer ID	Customer Income Level	Purchase Amount (USD)	\
0	1	2	0.250559	
1	2	1	0.431133	
2	3	0	0.118262	
3	4	2	0.127593	
4	5	2	0.365736	
	Frequency of Purchases	Age	Previous Purchases	Cluster
0	3	55	14	1
1	3	19	2	0
2	6	50	23	2
3	6	21	49	1
4	0	45	31	1

Figure 3.3.1.2 Result of K-means clustering

From the result, the cluster centres show the average value for each feature in every group. Cluster 0 has an average Customer Income Level of 0.2401 (indicating low-income), average Purchase Amount (USD) of 0.3045, and average Frequency of Purchases of 0.2110, which show customers who spend moderately but purchase infrequently. For Cluster 1, with average Customer Income Level of 1.0000 (indicating high-income), average Purchase Amount (USD) of 0.2989, and average Frequency of Purchases of 0.5005, represents high-income customers who make moderate purchases at a moderate frequency. Cluster 2 has an average Customer Income Level of 0.2524 (moderate-income), average Purchase Amount (USD) of 0.3042, and average Frequency of Purchases of 0.7799, which indicate the customers who make frequent purchases with moderate spending. The cluster labels show how customers are grouped according to their income level, frequency of purchases, and spending patterns. For example, a customer with income level 2 (high income), purchase amount 0.25 (moderate spending), frequency of purchase 3 (moderate purchase frequency) will be clustered into cluster 1, which represents high-income customers with moderate spending and moderate purchase frequency.

3.3.2 DBSCAN

DBSCAN Clustering: Impact of Customer Income Level and Other Features on Purchasing Behavior (eps=0.5, min_samples=5)

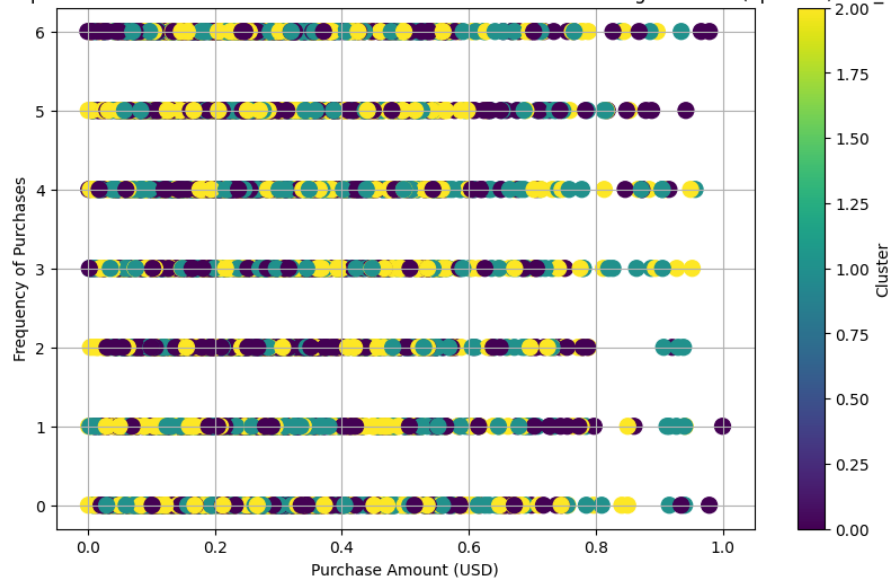


Figure 3.3.2.1 Scatter plot of DBSCAN clustering

From the scatter plot, we can observe the clusters based on Purchase Amount (USD) and Frequency of Purchases. The plot clearly shows how the clusters differ in terms of spending and purchase frequency. Cluster 0, represented by the purple colour, contains customers with low spending and infrequent purchases. Cluster 1, marked by green, groups customers with moderate spending and a moderate frequency of purchases. Finally, Cluster 2, in yellow, represents customers with relatively higher spending and frequent purchases. The plot visually captures the distinct purchasing behaviours of each cluster, highlighting how customers' income levels influence their spending patterns. The clustering was performed using DBSCAN with $\text{eps}=0.5$ and $\text{min_samples}=5$, where eps define the maximum distance between two points for them to be considered neighbours, and min_samples defines the minimum number of points required to form a dense region (cluster). These parameter values allowed DBSCAN to effectively identify clusters and exclude noise.

```

Number of points in each DBSCAN cluster (including noise points):
DBSCAN Cluster
0    1349
2    1295
1    1256
Name: count, dtype: int64

Number of noise points: 0

Average values for each cluster (for insight into cluster centers):
DBSCAN Cluster  Customer Income Level  Purchase Amount (USD) \
0                2.0                0.298916
1                1.0                0.307106
2                0.0                0.301606

DBSCAN Cluster  Frequency of Purchases  Age  Previous Purchases
0                3.002965  44.242402    24.892513
1                2.981688  43.343153    25.661624
2                2.938996  44.590734    25.528958

Dataset with DBSCAN Cluster Labels:
Customer ID  Customer Income Level  Purchase Amount (USD) \
0            1                    2    0.250559
1            2                    1    0.431133
2            3                    0    0.118262
3            4                    2    0.127593
4            5                    2    0.365736

Frequency of Purchases  Age  Previous Purchases  DBSCAN Cluster
0            3    55            14            0
1            3    19             2            1
2            6    50            23            2
3            6    21            49            0
4            0    45            31            0

```

Figure 3.3.2.2 Result of DBSCAN clustering

From the result, we can see the number of points in each DBSCAN cluster, where Cluster 0 has 1349 points, Cluster 1 has 1295 points, and Cluster 2 has 1256 points. There are 0 noise points, meaning all data points have been assigned to a cluster. The average values for each cluster show that Cluster 0 is characterized by high-income customers (average income level = 2.0) who have moderate spending (average purchase amount = 0.2989) and infrequent purchasing behaviour (average frequency of purchases = 3.0). Cluster 1 represents medium-income customers (average income level = 1.0) with moderate spending (average purchase amount = 0.3071) and moderate purchase frequency (average frequency of purchases = 2.98), while Cluster 2 consists of low-income customers (average income level = 0.0) who make frequent purchases (average frequency of purchases = 2.94) with moderate spending (average purchase amount = 0.3016). The dataset with the DBSCAN cluster labels shows how each customer is grouped based on their spending habits, providing useful insights into the relationship between income levels and purchasing behaviour. For example, a customer with income level 1 (medium income), purchase amount 0.43 (moderate spending), and frequency of purchase 3 (moderate purchase frequency) will be clustered into Cluster 0, which represents high-income customers with moderate spending and moderate purchase frequency.

4.0 Model Evaluation

Criteria	Description
Intra-cluster similarity	How close members of the same cluster are
Inter-cluster separation	How far apart the clusters are from each other
Compactness	Small average distance from members to centroid
Noise handling / Outlier detection	Ability to ignore irrelevant data
Shape detection	Ability to detect arbitrary shapes
Interpretability	Ease of understanding and using clusters for business
Davies-Bouldin Index (DBI)	Lower DBI = better clustering (only for K-Means)

4.0.1 Description Table

This section presents the evaluation and comparison of the two clustering models used in this analysis: K-Means and DBSCAN. Clustering quality should be assessed using internal validation techniques, such as intra-cluster similarity, inter-cluster separation, and compactness. When available, metrics like the Davies-Bouldin Index (DBI) further aid in assessing how well the clusters are separated and internally cohesive.

4.1 Age Group Dominance

Evaluation Metric	K-Means	DBSCAN
Cluster Count	3 clusters	7 clusters
Cluster Shape	Spherical	Arbitrary
Intra-cluster similarity	Good	Very high
Inter-cluster separation	Clear separation	lower inter-cluster separation
Outlier Handling	No	Effectively handles outliers
Interpretability	High – Younger, middle, older	Moderate interpretability

K-Means is the better model for age group segmentation due to its effective separation, low DBI, and business-usable groupings. DBSCAN, with 7 clusters, offers a simpler density - based grouping, useful if only broad age groupings are needed, but lacks the detailed breakdown of K-Means.

4.2 Payment Method

Evaluation Metric	K-Means	DBSCAN
Cluster Count	3	3
Cluster Shape	Spherical	Arbitrary
Intra-cluster similarity	Moderate to good	Moderate
Inter-cluster separation	Reasonable	Moderate
Outlier Handling	No	Yes
Interpretability	High	Moderate

In analysing payment method preferences, both K-Means and DBSCAN successfully formed three customer segments. However, K-Means produced more interpretable clusters aligned with business traits such as income level and purchase frequency. K-Means effectively identified low-income, low-frequency buyers; high-income, moderate buyers; and frequent but budget-conscious shoppers, making it ideal for targeted marketing strategies. DBSCAN also revealed similar patterns and handled data density well, but the behavioural distinctions among its clusters were less pronounced, and no noise points were detected. Overall, K-Means is the preferred model due to its clearer segmentation and better alignment with strategic decision-making.

4.3 Customer Income Level

Evaluation Metric	K-Means	DBSCAN
Cluster Count	3	3
DBI	Not reported	Not applicable
Cluster Shape	Spherical	Arbitrary
Intra-cluster similarity	Moderate	Moderate
Inter-cluster separation	Moderate	Moderate
Outlier Handling	No	Yes

In evaluating the impact of customer income level on purchasing behaviour, both K-Means and DBSCAN proved to be valid and effective clustering models. K-Means offered clear and interpretable cluster centroids, allowing straightforward identification of customer segments based on income, purchase amount, and purchase frequency, making it highly suitable for actionable business decisions. DBSCAN, while not centroid-based, successfully

grouped customers by density and was particularly effective in capturing natural clusters that may not follow linear boundaries. Although both models performed well, K-Means holds a slight advantage due to its simplicity, ease of interpretation, and practical applicability in business contexts.

5.0 Conclusion

References

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